📘 JavaScript Arrays & Array Methods

1.What is an Array in JavaScript?

An **Array** is a special object used to store multiple values in a single variable.

```
let numbers = [10, 20, 30, 40];
```

- Arrays are **zero-indexed**
- Can store **mixed data types**
- Dynamic (size can grow or shrink)

```
let mixed = [10, "Hello", true, { name: "John" }];
```

2.Creating Arrays

1. Literal Syntax (Recommended)

```
let fruits = ["Apple", "Banana", "Mango"];
```

2. Using Constructor

```
let numbers = new Array(1, 2, 3);
```

3.Accessing & Modifying Elements

```
let fruits = ["Apple", "Banana", "Mango"];


console.log(fruits[0]); // Apple


fruits[1] = "Orange";
console.log(fruits);
```

🔥 Core Array Methods

We will divide into categories:

1. Adding & Removing
2. Iteration
3. Transformation
4. Searching
5. Sorting
6. Other Important Methods

1.Adding & Removing Elements

- **push() – Add to End**

```
let nums = [1, 2, 3];
nums.push(4);

console.log(nums); // [1,2,3,4]
```

- **pop() – Remove from End**

```
let nums = [1, 2, 3];
nums.pop();

console.log(nums); // [1,2]
```

- **unshift() – Add to Beginning**

```
let nums = [2, 3];
nums.unshift(1);

console.log(nums); // [1,2,3]
```

- **shift() – Remove from Beginning**

```
let nums = [1, 2, 3];

nums.shift();


console.log(nums); // [2,3]
```

- **splice() – Add/Remove at Specific Position**

Syntax:

```
array.splice(start, deleteCount, item1, item2)
```

Remove

```
let nums = [1,2,3,4];

nums.splice(1,2);


console.log(nums); // [1,4]
```

Add

```
let nums = [1,4];

nums.splice(1,0,2,3);


console.log(nums); // [1,2,3,4]
```

2.Iteration Methods

- **forEach()**

Used for looping.

```
let nums = [10,20,30];
```

```
nums.forEach(function(n){
  console.log(n * 2);
});
```

Arrow Function Version:

```
nums.forEach(n => console.log(n));
```

3.Transformation Methods

These return **new arrays**

- **map()**

Transforms each element.

```
let nums = [1,2,3];

let squared = nums.map(n => n * n);

console.log(squared); // [1,4,9]
```

- **filter()**

Filters elements based on condition.

```
let nums = [10,15,20,25];

let even = nums.filter(n => n % 2 === 0);

console.log(even); // [10,20]
```

- **reduce()**

Reduces array to single value.

```
let nums = [1,2,3,4];

let sum = nums.reduce((total, n) => total + n, 0);

console.log(sum); // 10
```

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4. **Searching Methods**

- **find()**

Returns first matching element.

```
let users = [
  {id:1, name:"John"},
  {id:2, name:"Jane"}
];

let user = users.find(u => u.id === 2);

console.log(user);
```

- **findIndex()**

```
let nums = [5,10,15];

let index = nums.findIndex(n => n === 10);
```

```
console.log(index); // 1
```

- **includes()**

```
let fruits = ["Apple","Banana"];

console.log(fruits.includes("Apple")); // true
```

- **indexOf()**

```
let nums = [10,20,30];

console.log(nums.indexOf(20)); // 1
```

5.Sorting Methods

- **sort()**

⚠ Default sorts as string!

```
let nums = [5,2,8,1];

nums.sort((a,b) => a - b);

console.log(nums); // [1,2,5,8]
```

Descending:

```
nums.sort((a,b) => b - a);
```

- **reverse()**

```
let nums = [1,2,3];
```

```
nums.reverse();

console.log(nums); // [3,2,1]
```

6.Other Important Methods

- **concat()**

```
let a = [1,2];
let b = [3,4];

let result = a.concat(b);

console.log(result); // [1,2,3,4]
```

- **slice()**

```
let nums = [1,2,3,4,5];

let part = nums.slice(1,4);

console.log(part); // [2,3,4]
```

- **join()**

```
let fruits = ["Apple","Banana"];

console.log(fruits.join(", "));
```

// Apple, Banana

⚡ **Summary Table**

Method	Purpose	Returns New Array?
push	Add end	❌
pop	Remove end	❌
map	Transform	✅
filter	Filter	✅
reduce	Aggregate	❌ (returns value)
find	First match	❌
sort	Sort	❌
slice	Extract	✅